

Ran Wang

Scientific Mobility | Firm Innovation | Regional Innovation | Longitudinal Relational Data
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RESEARCH PROFILE

Master's candidate in Human Geography at East China Normal University, trained in economic geography, regional innovation, and quantitative spatial analysis. Research focuses on high-skilled scientific mobility, collaboration networks, destination choice, city networks, firm innovation, and innovation measurement. The work uses bibliometric records, public career histories, institution-city linkage, longitudinal relational data, and reproducible Python/R workflows to connect individual scientific careers, firm innovation records, and urban-regional knowledge structures.

EDUCATION

East China Normal University, School of Geographic Sciences

M.Sc. in Human Geography, 2023-2026

- Research field: economic geography and regional innovation; master's thesis on intercity mobility networks of top global biomedical scientists.
- Relevant training: relational economic geography, innovation economics, frontiers of human geography, and interdisciplinary research methods.

Northeast Normal University, School of Geographical Sciences

B.Sc. in Human Geography and Urban-Rural Planning, 2019-2023

- Completed coursework and training in quantitative geography, GIS spatial analysis, urban geography, regional analysis and planning, and urban planning methods.

PUBLICATIONS AND MANUSCRIPTS

城市网络视角下全球生物医药顶尖科学家流动的空间结构与影响因素

Accepted, World Regional Studies

- Role: First author.
- Constructed career-history and intercity mobility data for global top biomedical scientists and analyzed city mobility networks with social network analysis.
- Examined mobility-tie formation and mobility intensity across career stages from an urban-network perspective.

“全域—自我”视角下全球生物医药顶尖科学家城际流动网络结构与形成机制

Submitted Manuscript

- Role: First author.
- Examines intercity mobility networks of elite biomedical scientists from whole-network and ego-network perspectives.
- Integrates social network analysis, mobility ego networks, multidimensional proximity, historical mobility ties, and cross-city and cross-platform career trajectories.

"Pre-move coauthor ties and the direction of scientific mobility: A city-level choice analysis of highly cited biomedical scientists."

Submitted Manuscript

- Role: First author.
- Organizes observed intercity employment moves as event-specific destination choices and compares selected cities with alternative destinations.
- Measures destination-specific direct coauthor ties and related timing structure before the move, interpreting results as associations with destination choice.

Wang, Ran. "Intercity Mobility Network of Top Global Biomedical Scientists: A Dual Perspective of Whole and Ego Networks."

Master's Thesis, 2026

- Used a 500-scientist, 1970-2023 longitudinal database to connect whole-network analysis, mobility ego networks, WERGM, GIS, and career-trajectory classification.

SELECTED RESEARCH EXPERIENCE

Global Mobility and Collaboration Networks of Elite Biomedical Scientists

2023-present

- Constructed a longitudinal research database covering 500 scientists, 1970-2023 career histories, appointments, publication locations, collaboration records, mobility events, and city attributes.
- Developed scientist-institution-city linkage, directed weighted mobility networks, WERGM specifications, mobility ego-network measures, and Python/R quality-control workflows.

Destination Choice and Pre-move Collaboration

2024-present

- Constructed event-specific candidate-city sets and measured destination-specific direct coauthor ties for mobility choice analysis.
- Estimated event-stratified conditional logit models with clustered robust inference, timing and persistence measures, and robustness checks.

Scopus Research Data Toolkit / Research Data Infrastructure

2024-present

- Implemented author-publication linkage, author-affiliation linkage, institution and city normalization, city-year controls, and reproducible table-building workflows.
- Built synthetic fixtures, offline mock transport, quality audits, and automated tests to support repeatable bibliometric data processing.

Technological and Brand Innovation among Leading Global R and D Firms

2024

- Assembled firm-year data for the top 2,000 corporate R and D investors, integrating R and D expenditure, patents, trademarks, industry, and country information.
- Prepared panel data for analyses of technological innovation, brand innovation, and cross-national firm heterogeneity.

Daily Nighttime Lights and Urban Resilience

2023

- Processed VIIRS-DNB daily nighttime lights for 48 cities and applied Fisher Information Light, dynamic time warping, and k-means clustering.

Global Trade Network Evolution during COVID-19

2022

- Used UN Comtrade data from 2019-2021 to construct directed weighted trade networks and analyze changes in centrality and core structure.

POLICY AND PLANNING EXPERIENCE

China Development Institute, Shenzhen

Research Assistant / Intern

- Contributed to research on port planning, regional development, and industrial spatial organization through policy and literature review, case collection, data organization, and report drafting.
- Participated in territorial spatial planning and county industrial planning tasks, including GIS mapping, document preparation, and comparative policy synthesis.

QUANTITATIVE MODELLING EXPERIENCE

Mathematical Modelling and Predictive Analysis

- Applied Lasso, recursive feature elimination, random forest, SVM, grid search, and K-fold cross-validation in mathematical modeling competition projects.
- Estimated OLS and Steinmetz-equation models for empirical prediction and parameter comparison tasks.

METHODS AND TECHNICAL SKILLS

Empirical Methods

- Conditional logit, Heckman two-stage model, WERGM, clustered robust inference, robustness checks, and longitudinal data construction.

Network and Spatial Methods

- Social network analysis, directed weighted networks, ego networks, centrality analysis, GIS mapping, city normalization, and spatial data integration.

Programming and Data

- Python, R, Stata, ArcGIS/QGIS, SQL-style data cleaning, version-controlled workflows, reproducible scripts, quality audits, and automated tests.

Research Databases

- Scopus, OpenAlex, UN Comtrade, patent and trademark data, corporate R&D; data, VIIRS-DNB nighttime lights, and public institutional records.

Research Communication

- Academic writing in Chinese and English, manuscript preparation, literature synthesis, research report writing, table and figure preparation, and presentation design.

HONORS

Academic and Competition Honors

- Outstanding Student, East China Normal University.
- Academic Scholarship, East China Normal University.
- Third-Class Scholarship, Northeast Normal University.
- Innovation and Entrepreneurship Scholarship, Northeast Normal University.
- Excellent completion, National College Student Innovation and Entrepreneurship Training Program; ranked first and served as project leader.